The origin of ferromagnetism and the Hubbard model

appendix 2: positivity and uniqueness of the ground state of a Schrödinger equation

I used ChatGPT 5.4 Pro to prepare this lecture

Advanced Topics in Statistical Physics
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§ Setting and main theorem

dimension $D = 1, 2, 3, \ldots$ ($D = 3N$ for N particle system in three dim.)

bounded open connected domain $\Omega \subset \mathbb{R}^D$ e.g. $\Omega = (0, L)^D$

$\partial\Omega$: boundary of Ω closure $\bar{\Omega} = \Omega \cup \partial\Omega$ (1)

$\mathbf{r} = (x_1, \ldots, x_D) \in \Omega$ (2) $\Delta = \sum_{j=1}^{D} \frac{\partial^2}{\partial x_j^2}$ (3)

potential $V(\mathbf{r})$ real, continuous on $\bar{\Omega}$ and hence bounded → essential

may be relaxed easily

Schrödinger equation and boundary condition

$$\begin{cases} \{-\Delta + V(\mathbf{r})\}\varphi(\mathbf{r}) = E\,\varphi(\mathbf{r}) \quad \mathbf{r} \in \Omega & (4) \\ \varphi(\mathbf{r}) \text{ extends continuously to } \bar{\Omega} \text{ and } \varphi(\mathbf{r}) = 0 \text{ if } \mathbf{r} \in \partial\Omega \end{cases}$$

$$\left(\hat{H}|\varphi\rangle = E|\varphi\rangle \quad (5) \quad \text{with} \quad \hat{H} = -\Delta + V(\hat{\mathbf{r}}) \quad (6) \right)$$

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theorem the ground state of Sch. eq. P1-(4) is unique (nondegenerate) and the corresponding eigenfunction $\varphi(\cdot)$ can be chosen so that $\varphi(\mathfrak{r}) > 0$ for any $\mathfrak{r} \in \Omega$.

"ground state wave function is nodeless"

continuum analogue of Perron-Frobenius theorem

we shall present a "physcists' proof" without functional analysis, but the proof may be made into a math proof with extra care

(the theorem and its (sophisticated) proof can be found in many mathematical textbooks on QM, such as

Gerald Teschl "Mathematical Methods in Quantum Mechanics" §10.5

you can legally download PDF from author's web page!)

§ nonnegative eigenfunction is positive

lemma 1 suppose that $\varphi(\cdot)$ satisfies Sch. eq. P1-(4) for some E, and $\varphi(\mathbf{r}) \geq 0$ for all $\mathbf{r} \in \Omega$.

then $\varphi(\mathbf{r}) > 0$ for all $\mathbf{r} \in \Omega$ or $\varphi(\mathbf{r}) = 0$ for all $\mathbf{r} \in \Omega$

proof we shall prove

☆ if $\varphi(\mathbf{r}_0) = 0$ for some $\mathbf{r}_0 \in \Omega$ then $\varphi(\mathbf{r}) = 0$ for all $\mathbf{r} \in \Omega$.

write Sch. eq. P1-(4) as

$$\Delta \varphi(\mathbf{r}) = U(\mathbf{r})\, \varphi(\mathbf{r}), \quad \mathbf{r} \in \Omega \qquad (1)$$

$$\text{with } U(\mathbf{r}) = V(\mathbf{r}) - E \qquad (2)$$

there is U_{max} such that $U(\mathbf{r}) \leq U_{max}$ for any $\mathbf{r} \in \overline{\Omega}$ (3)

(iii) The case with $D=1$ (easy)

$\Omega = (a, b)$. suppose $\varphi(x_0) = 0$ at some $x_0 \in \Omega$.

since $\varphi(x) \geq 0$, we have $\varphi'(x_0) = 0$

now the Sch. eq. p3-(1) reads ODE!

$$\varphi''(x) = V(x)\,\varphi(x) \qquad (1)$$

$\varphi(x) = 0$ for all $x \in (a, b)$ is a solution of ODE (1) with "initial conditions" $\varphi(x_0) = \varphi'(x_0) = 0$. $\quad (2)$

from the uniqueness of the solution of ODE (1), we get ☆

(iii) the case with general D

for $\ell > 0$, sphere $S(\ell) = \{\mathbf{r} \in \mathbb{R}^D \mid |\mathbf{r}| = \ell\}$ (1)

ball $B(\ell) = \{\mathbf{r} \in \mathbb{R}^D \mid |\mathbf{r}| \leq \ell\}$ (2)

volume $|B(\ell)|$ $\qquad \frac{d}{d\ell}|B(\ell)| = |S(\ell)|$ (3) $\leftarrow$ surface area

assume $\varphi(\mathbf{r}_0) = 0$ for some $\mathbf{r}_0 \in \Omega$

$\mathbf{r} = \ell \mathbf{s}$

for $\ell \geq 0$ $\quad m(\ell) := \frac{1}{|S(\ell)|}\int_{\mathbf{r} \in S(\ell)} da(\mathbf{r})\, \varphi(\mathbf{r}_0 + \mathbf{r}) = \frac{1}{|S(1)|}\int_{\mathbf{s} \in S(1)} da(\mathbf{s})\, \varphi(\mathbf{r}_0 + \ell \mathbf{s})$ (3)

surface average

$da(\cdot)$ scalar surface element
$d\mathbf{a}(\cdot)$ vector surface element

$$\frac{\partial}{\partial \ell}\varphi(\mathbf{r}_0 + \ell \mathbf{s}) = \mathbf{s} \cdot \operatorname{grad} \varphi(\mathbf{r}_0 + \ell \mathbf{s}) \quad (4)$$

$$= \sum_{j=1}^{D} s_j \frac{\partial}{\partial x_j}\varphi(\mathbf{r})\Big|_{\mathbf{r} = \mathbf{r}_0 + \ell \mathbf{s}}$$

$$\mathbf{s}\, da(\mathbf{s}) = d\mathbf{a}(\mathbf{s}) \quad (5)$$

$$m'(l) = \frac{1}{|S(1)|}\int_{s\in S(1)} da(s)\cdot \mathrm{grad}\,\varphi(r_0 + l s) = \frac{1}{|S(l)|}\int_{r\in S(l)} da(r)\cdot \mathrm{grad}\,\varphi(r_0 + r)$$

Gauss' th.

$$= \frac{1}{|S(l)|}\int_{r\in B(l)} \underbrace{dr}_{\prod_{j=1}^{D} dx_j \text{ volume element}}\; \underbrace{\mathrm{div}\,\mathrm{grad}\,\varphi(r_0+r)}_{\Delta\varphi(r_0+r)} = \frac{1}{|S(l)|}\int_{r\in B(l)} dr\, U(r_0+r)\,\varphi(r_0+r)$$

$\varphi(r)\geq 0$

$$\leq \frac{U_{max}}{|S(l)|}\int_{r\in B(l)} dr\, \varphi(r_0+r) = \frac{U_{max}}{|S(l)|}\int_0^l dq \underbrace{\int_{r\in S(q)} da(r)\,\varphi(r_0+r)}_{=|S(q)|\, m(q)} \qquad (1)$$

$$\boxed{\mu(l) := \max_{q\in[0,l]} m(q)} \qquad (2)$$

$$|B(l)| = \frac{l}{D}|S(l)| \leftarrow \left(\begin{array}{l} |B(l)| = A l^D \\ |S(l)| = \frac{d}{dl}|B(l)| = D A l^{D-1} \end{array}\right)$$

$$m'(l) \leq \frac{U_{max}}{|S(l)|}\mu(l) \overbrace{\int_0^l dq\, |S(q)|} = \frac{U_{max}\, l}{D}\mu(l) \qquad (3)$$

for $s\in[0,l]$

$$m'(s) \leq \frac{U_{max}\, s}{D}\mu(s) \leq \frac{U_{max}\, s}{D}\mu(l) \qquad (4)$$

for $s \in [0, \ell]$ $\quad m'(s) \leq \frac{U_{max}\, s}{D}\mu(\ell)$ (1)

$m(0) = 0$ (2) $\rightarrow$ $m(q) = \int_0^q ds\, m'(s) \leq \frac{U_{max}\, q^2}{2D}\mu(\ell)$ (3)

for $q \in [0, \ell]$

$$\left(\mu(\ell) = \max_{q \in [0,\ell]} m(q) = \max_{q\in[0,\ell]} \frac{1}{|S(q)|}\int_{r \in S(q)} da(r)\, \varphi(r_0 + r) \quad (4)\right)$$

$$\mu(\ell) \leq \frac{U_{max}\, \ell^2}{2D}\mu(\ell) \quad (5)$$

if $\frac{U_{max}\, \ell^2}{2D} < 1$ (6) this is possible only when $\mu(\ell) = 0$ (7)

we showed that

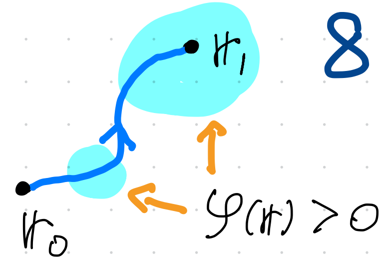
$$\boxed{\begin{gathered}\varphi(r_0) = 0 \quad \text{for } r_0 \in \Omega \\ \Downarrow \\ \varphi(r) = 0 \quad \text{for any } r \text{ s.t. } |r - r_0| < \sqrt{\frac{2D}{U_{max}}}\end{gathered}}$$

☆☆

we are almost done!

Suppose $\varphi(x_1) > 0$ for $x_1 \in \Omega$

x_t, $t \in [0,1]$ a continuous path in Ω



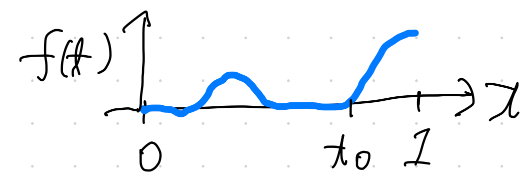
$\varphi(x)$ is continuous

(1)

$$f(t) = \varphi(x_t) \text{ is continuous in } t \in [0,1]$$

$$f(0) = 0,\ f(1) > 0 \qquad (2)$$

there is t_0 s.t. $f(t_0) = 0$ and $f(t) > 0$ for $t \in (t_0, 1]$ (3)



this contradicts ☆☆ if we use x_{t_0} as x_0

☆ in p.3 is proved

proof of lemma 1 in appendix 1

(a standard cool proof (see, e.g., Teschl) $\alpha > 0$

if $\varphi(x) \geq 0$ for $x \in \Omega$ and $\|\varphi\| \neq 0$ then $(e^{-\alpha \hat{H}} \varphi)(x) > 0$ for $x \in \Omega$)

§ proof of the main theorem

essentially the same as in the proof of Perron-Frobenius theorem → especially from our standard of rigor

lemma 2 it suffices to consider only energy eigenfunctions (i.e., $\varphi(\cdot)$ satisfying Sch. eq. P.1-(4)) such that $\varphi(\mathbf{r}) \in \mathbb{R}$ for all $\mathbf{r} \in \Omega$

proof

$$\{-\Delta + V(\mathbf{r})\}\varphi(\mathbf{r}) = E\,\varphi(\mathbf{r}) \qquad (1)$$

$$\Updownarrow$$

$$\{-\Delta + V(\mathbf{r})\}\varphi^*(\mathbf{r}) = E\,\varphi^*(\mathbf{r}) \qquad (2)$$

$\varphi^*(\mathbf{r})$ is also an eigenfunction

at least one of $\frac{1}{2}\left(\varphi(\mathbf{r}) + \varphi^*(\mathbf{r})\right)$ and $\frac{1}{2i}\left(\varphi(\mathbf{r}) - \varphi^*(\mathbf{r})\right)$ is nonzero //

notation wave functions $\varphi(\cdot)$, $\psi(\cdot)$

inner product $$\langle \varphi, \psi \rangle = \int_{\mathbf{r}\in\Omega} d\mathbf{r}\; \varphi^*(\mathbf{r})\,\psi(\mathbf{r}) \qquad (3)$$

lemma 3 (variational principle) E_{GS} : the ground state energy

$$\langle \varphi, (\hat{H}-E_{GS})\varphi \rangle \geq 0 \quad \text{for any } \varphi(\cdot) \qquad (1)$$

$$\langle \varphi, (\hat{H}-E_{GS})\varphi \rangle = 0 \iff \{-\Delta + V(x)\}\varphi(x) = E_{GS}\varphi(x) \qquad (2)$$

"proof" $\hat{H}$ is self-adjoint, and the spectrum consists only of eigenvalues.

Ω is bounded

→ the set of energy eigen states $\{\psi_j(\cdot)\}_{j=0,1,2,\ldots}$

form a complete orthonormal system

lemma follows by expanding $\varphi(\cdot) = \sum_{j=0}^{\infty} \alpha_j \psi_j(\cdot)$ (4)

we also note

$$\langle \varphi, (\hat{H}-E_{GS})\varphi \rangle = \int dx\, \varphi^*(x)\{-\Delta + \underbrace{V(x)-E_{GS}}_{=U_0(x)}\}\varphi(x)$$

$$= \int dx \{ \underbrace{|\mathrm{grad}\,\varphi(x)|^2}_{\sum_{j=1}^{D}\left|\frac{\partial}{\partial x_j}\varphi(x)\right|^2} + U_0(x)|\varphi(x)|^2 \} \qquad (5)$$

proof of the main theorem

let $\varphi(\cdot)$ be a ground state, i.e., $\{-\Delta + V(\mathbf{r})\}\varphi(\mathbf{r}) = E_{GS}\varphi(\mathbf{r})$ (1)

we can assume $\varphi(\mathbf{r}) \in \mathbb{R}$ and $\varphi(\mathbf{r}) > 0$ for some $\mathbf{r} \in \Omega$.

decompose $\varphi(\cdot) = \varphi_+(\cdot) + \varphi_-(\cdot)$ (2)

with
$$\begin{cases} \varphi_+(\mathbf{r}) = \max\{\varphi(\mathbf{r}), 0\} & (3) \\ \varphi_-(\mathbf{r}) = \min\{\varphi(\mathbf{r}), 0\} & (4) \end{cases}$$

$$\underbrace{\langle \varphi, (\hat{H} - E_{GS})\varphi \rangle}_{=\,0} = \int_{\mathbf{r}\in\Omega} d\mathbf{r}\, \Big\{ |\mathrm{grad}\,\varphi(\mathbf{r})|^2 + \overbrace{v_0(\mathbf{r})}^{V(\mathbf{r}) - E_{GS}} |\varphi(\mathbf{r})|^2 \Big\}$$

$$= \int d\mathbf{r}\, \Big\{ |\mathrm{grad}\,\varphi_+(\mathbf{r})|^2 + v_0(\mathbf{r})(\varphi_+(\mathbf{r}))^2 \Big\}$$

$$+ \int d\mathbf{r}\, \Big\{ |\mathrm{grad}\,\varphi_-(\mathbf{r})|^2 + v_0(\mathbf{r})(\varphi_-(\mathbf{r}))^2 \Big\}$$

$$= \underbrace{\langle \varphi_+, (\hat{H} - E_{GS})\varphi_+ \rangle}_{\geq 0} + \underbrace{\langle \varphi_-, (\hat{H} - E_{GS})\varphi_- \rangle}_{\geq 0} \quad (5)$$

we have $\langle \varphi_+, (\hat{H} - E_{GS}) \varphi_+ \rangle = 0$ (1)

and hence $\{-\Delta + V(\mathbf{r})\} \varphi_+(\mathbf{r}) = E_{GS}\, \varphi_+(\mathbf{r})$ (2) $(\mathbf{r} \in \Omega)$

since $\varphi_+(\mathbf{r}) \geq 0$ for all $\mathbf{r} \in \Omega$ and $\varphi_+(\mathbf{r}) > 0$ for some $\mathbf{r} \in \Omega$,

we see $\varphi_+(\mathbf{r}) > 0$ for all $\mathbf{r} \in \Omega$ and hence

$$\varphi(\mathbf{r}) = \varphi_+(\mathbf{r}) > 0 \text{ for all } \mathbf{r} \in \Omega$$

(ii) uniqueness

suppose there are two ground states, $\varphi_1(\cdot)$ and $\varphi_2(\cdot)$.

we can chose them so that $\varphi_1(\mathbf{r}) > 0$, $\varphi_2(\mathbf{r}) > 0$ for all $\mathbf{r} \in \Omega$

fix $\mathbf{r}_0 \in \Omega$ and set $\psi(\mathbf{r}) = \varphi_1(\mathbf{r}) - \dfrac{\varphi_1(\mathbf{r}_0)}{\varphi_2(\mathbf{r}_0)} \varphi_2(\mathbf{r})$ for $\mathbf{r} \in \Omega$

we have $\{-\Delta + V(\mathbf{r})\} \psi(\mathbf{r}) = E_{GS}\, \psi(\mathbf{r})$

$\psi(\mathbf{r}) \in \mathbb{R}$ for all $\mathbf{r} \in \Omega$ and $\psi(\mathbf{r}_0) = 0$.

it is only possible that $\psi(\mathbf{r}) = 0$ for all $\mathbf{r} \in \Omega$ //